Ayush Kumar Jakhmola

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Summary

Offensive Security Researcher and Penetration Tester focused on web application security, network penetration testing, vulnerability assessment, and security research. Interested in cloud security, Wi-Fi security, Active Directory, and AI/LLM application security. Hands-on experience with cybersecurity tools, Linux environments, security labs, and security-focused project development.

Education

BCA (Hons.) Cyber Security — Graphic Era Hill University, Dehradun 2024 - 2027

Senior Secondary Education — S.G.R.R. Public School, Kotdwara 2023 - 2024

High School — S.G.R.R. Public School, Kotdwara 2021 - 2022

Skills

Security Domains: Web Application Security, Network Penetration Testing, Vulnerability Assessment, Cloud Security, Wi-Fi Security, Active Directory, AI/LLM Security
Operating Systems: Kali Linux, Ubuntu Server, Windows 10/11, Linux CLI, Bash, SSH, System Administration
Linux & Systems: Linux CLI, Bash, system services, processes, permissions, virtualization, system troubleshooting
Networking: TCP/IP, DNS, HTTP/HTTPS, IP addressing, subnetting, network troubleshooting, REST APIs, Wireshark, Nmap
Cloud Infrastructure: AWS EC2, VPC, IAM, S3,
Programming & Scripting: C, Python, Bash, CMD
Tools & Platforms: Git, GitHub, GitLab, SSH, VS Code, Postman
Soft Skills: Team Collaboration, Communication, Leadership, Problem Solving

Projects

Secure Cloud Online Banking System

- Developed a full-stack banking web application featuring user authentication, email OTP verification, account management, and transaction history.
- Implemented banking workflows for deposits, withdrawals, and account-to-account transfers, along with an administrative dashboard for user, account, and transaction monitoring.
- Explored application security controls including Bcrypt password hashing, AES-GCM encryption utilities, CSRF protection, and session-based access checks.

Password Cracking & Analyzer Tool

- Developed a Flask-based web application to analyze password strength using five criteria: length, uppercase, lowercase, digits, and symbols.
- Implemented theoretical entropy and estimated cracking-time calculations to demonstrate password resilience against brute-force attacks.
- Built an interactive cracking simulation comparing Linear Search and Binary Search, with attempt counts and execution-time metrics to illustrate algorithmic efficiency.

IoT-Based Weather Monitoring System | ESP8266, Embedded C, Wi-Fi, ThingSpeak, Sensors

- Built an ESP8266-based monitoring system integrating DHT11, BMP180, OLED display, LEDs, and buzzer for real-time environmental monitoring.
- Enabled up to 35% faster detection of abnormal temperature and humidity conditions through continuous sensor-based monitoring and remote visualization.
- Integrated sensor interfaces, hardware components, embedded software, and cloud connectivity for continuous monitoring applicable to equipment/server-room environments.

Certifications

CCNA: Introduction to Networks
Networking Basics (Cisco)
Network Defense
AWS Cloud Practitioner Essentials
Ethical Hacker (Cisco)
Introduction to CyberSecurity (Cisco)
Cyber Security Analyst Job Simulation
More Certifications